PROJECT PROPOSAL

Noise-Augmented Urban Sound Classification under Matched and Mismatched Conditions

A controlled study of whether robustness learned from one noise condition transfers to unseen noise

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Repository	https://github.com/tongtongchen-540/elec5305-project-540334953		
Project site	https://tongtongchen-540.github.io/elec5305-project-540334953/		

1. Project Overview

Urban sound classifiers recognise events such as sirens, dog barks, and engines. Deployment audio may contain traffic, crowds, wind, or sensor noise absent during training; this is a noise mismatch. The project will reproduce an MFCC-SVM classifier on UrbanSound8K, then add selected noises to training waveforms. Clean, single-noise, and mixed-noise training will be tested on familiar and unseen noises. Rather than claiming a new classifier, the study measures whether simple augmentation generalises or merely overfits to its training noise.

2. Background and Literature Rationale

UrbanSound8K contains 8,732 labelled excerpts of up to four seconds from ten urban sound classes. Its predefined ten folds keep excerpts derived from the same source recording together, avoiding an inflated result caused by train-test leakage [1]. Salamon et al. also reported that their MFCC baseline was sensitive to background interference. MFCCs summarise short-time spectral envelopes on a perceptual frequency scale and provide a compact, interpretable representation [2]. Additive noise, however, changes spectral energy and can shift the coefficients. Dennis et al. showed substantial performance loss under mismatched noise and proposed a robust spectrogram-image feature [3]. That method is more specialised and complex than the reproducible MFCC baseline selected here.

Salamon and Bello found that waveform augmentation improved a convolutional environmental-sound model, but the benefit varied by sound class and augmentation type [4]. Their CNN can learn temporal patterns that summary MFCC statistics may miss, but it requires more computation and complicates causal comparison. This study holds the feature-classifier pipeline fixed and varies only training noise. Real noises will come from DEMAND [5]. The matched-versus-mismatched comparison extends prior work by testing whether augmentation transfers to a noise environment absent from training.

3. Research Question and Contribution

The single research question is: To what extent does waveform-level noise augmentation improve an MFCC-SVM urban sound classifier when test noise is matched to, or different from, the noise used during training? The project will:

- reproduce a clean-audio MFCC-SVM baseline using the official folds;
- compare clean, white-noise, traffic-noise, and mixed-noise training strategies;
- separate matched tests from mismatched and completely held-out noise tests; and
- explain accuracy changes by noise type, SNR, and sound class.

4. Proposed Methodology



4.1 Data and preprocessing

UrbanSound8K audio and metadata will be processed in Python using librosa, NumPy, pandas, and scikit-learn. Clips will be converted to mono, resampled to 22.05 kHz, and padded or cropped only where required for consistent processing. The official ten-fold protocol will be preserved: nine folds train the model and the remaining fold tests it, repeated ten times. Noise recordings and waveform excerpts will also be separated before mixing, so a test mixture never reuses a training event or training noise segment. All scaling and model selection will use training data only.

4.2 Features and baseline model

Each clip will be divided into overlapping short frames and transformed with the STFT. A 40-band log-Mel spectrogram will be computed, followed by 20 MFCCs plus first- and second-order temporal differences (delta and delta-delta coefficients). Frame-level features will be summarized by their mean and standard deviation to form one fixed-length vector per clip. A standardised radial-basis-function SVM is selected because it handles nonlinear class boundaries, works well with medium-sized feature sets, and is practical on a personal computer. Its weakness is that temporal order is compressed by summary statistics; this trade-off will be acknowledged when results are compared with CNN literature.

4.3 Matched and mismatched noise experiment

Four systems will share exactly the same MFCC-SVM pipeline: clean training, white-noise training, DEMAND traffic-noise training, and mixed training using white, pink, and traffic noise. Training copies will be mixed at 20, 10, or 0 dB SNR. Every system will then be evaluated on clean audio and on white, pink, traffic, and held-out crowd noise at the same three SNRs. White-to-white and traffic-to-traffic are matched conditions; other pairings are mismatched, while crowd noise is held out from all training. The unchanged architecture and equalised training-set sizes make noise exposure the primary experimental variable. Fixed random seeds and saved mixture metadata will make all comparisons repeatable.

4.4 Evaluation and analysis

The primary metrics will be mean classification accuracy and macro F1-score across the ten official folds, with standard deviation and paired fold-wise differences. Macro F1 gives each class equal weight. An accuracy-versus-SNR curve will show degradation, and a training-noise by test-noise heatmap will expose transfer patterns. Confusion matrices and per-class recall will identify class-specific effects. The main quantities of interest are the gain over clean training, the gap between matched and mismatched tests, and any loss of clean-condition accuracy. Runtime and feature dimensionality will also be reported.

5. Expected Outcomes

The project will deliver working Python code, saved configurations and results, a report, and a GitHub Pages demonstration with audio examples. Mixed-noise training is expected to be less specialised but more stable on unseen noise, whereas a single-noise model may perform best when its training and test conditions match. A null or contrary result will still answer the question by revealing limits of simple augmentation. The discussion will relate errors to spectral masking, MFCC sensitivity, class similarity, and the strengths and weaknesses identified in the literature.

6. Feasibility and Scope Control

The core system uses standard libraries and a modest classical model, so it can run without specialised hardware. Dataset size and repeated experiments will be controlled by caching features, piloting on two folds, fixing the noise/SNR grid in advance, and keeping neural-network comparisons within the literature review rather than expanding the implementation scope.

7. Timeline (Weeks 1-13)

Weeks	Planned work	Milestone
1-2	Confirm topic, research question, repository, and proposal	Approved proposal
3-5	Literature review; download and audit UrbanSound8K; build preprocessing and MFCC extraction	Validated dataset pipeline
6-7	Implement and validate the clean MFCC-SVM baseline using the official folds	Reproduced baseline
8-9	Create matched/mismatched mixtures; implement single-noise and mixed-noise training	Modified systems
10-11	Run the full experiment; create heatmaps, SNR curves, confusion matrices, and error analysis	Final measurements
12-13	Prepare report, GitHub Pages documentation, audio demonstration, and presentation video	Final submission

8. References

[1] J. Salamon, C. Jacoby, and J. P. Bello, "A Dataset and Taxonomy for Urban Sound Research," Proceedings of the 22nd ACM International Conference on Multimedia, pp. 1041-1044, 2014. doi:10.1145/2647868.2655045.

[2] S. B. Davis and P. Mermelstein, "Comparison of Parametric Representations for Monosyllabic Word Recognition in Continuously Spoken Sentences," IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. 28, no. 4, pp. 357-366, 1980. doi:10.1109/TASSP.1980.1163420.

[3] J. Dennis, H. D. Tran, and H. Li, "Spectrogram Image Feature for Sound Event Classification in Mismatched Conditions," IEEE Signal Processing Letters, vol. 18, no. 2, pp. 130-133, 2011. doi:10.1109/LSP.2010.2100380.

[4] J. Salamon and J. P. Bello, "Deep Convolutional Neural Networks and Data Augmentation for Environmental Sound Classification," IEEE Signal Processing Letters, vol. 24, no. 3, pp. 279-283, 2017. doi:10.1109/LSP.2017.2657381.

[5] J. Thiemann, N. Ito, and E. Vincent, "The Diverse Environments Multi-channel Acoustic Noise Database (DEMAND)," Proceedings of Meetings on Acoustics, vol. 19, 035081, 2013. doi:10.1121/1.4806631.

Submission note	The proposal PDF will be uploaded to the public repository above. The linked GitHub Pages site will contain the project description, references, implementation progress, results, and audio demonstration.